Comparison of Machine Learning Algorithms and Large Language Models for Product Categorization

Abdullah İhsanoğlu
Özyeğin University
İstanbul, Türkiye
abdullah.ihsanoglu@ozu.edu.tr

Mounes Zaval
Özyeğin University, Huawei Türkiye R&D Center
İstanbul, Türkiye
mounes.zaval@{ozu.edu.tr,huawei.com}

Olcay Taner Yıldız
Özyegin University
İstanbul, Türkiye
olcay.yildiz@ozyegin.edu.tr

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***Özetçe* —Bu çalışma, geleneksel makine öğrenimi algoritmaları ve Büyük Dil Modelleri (LLM'ler) ile e-ticaret platformları için ürün kategorizasyonunun otomatikleştirilmesindeki etkinliği araştırmaktadır. Bu metodolojileri karşılaştırarak, çeşitli ürün listelerini sınıflandırma performanslarını değerlendiriyoruz. Bulgularımız, bu bağlamda LLM'lerin, karmaşık metin verilerini anlama ve kategorize etme konusunda geleneksel makine öğrenimi teknikleriyle benzer performans gösterdiğini belirtmektedir, bu da LLM'lerin bu bağlamda kullanımının gereksiz olabileceğini öne sürmektedir. Sonuçta, tercihin her modelin işletme maliyetleri ve kaynak tüketimine bağlı olduğu anlaşılmaktadır. Bu çalışma, hızla genişleyen çevrimiçi pazar yerleri bağlamında mevcut metin kategorizasyon tekniklerinin yetenekleri ve sınırlılıkları hakkında içgörüler sağlayarak alana katkıda bulunmaktadır.**

***Abstract*—This study explores the efficacy of traditional machine learning algorithms and Large Language Models (LLMs) in automating product categorization for online e-commerce platforms. By comparing these methodologies, we assess their performance in classifying a diverse range of product listings. Our findings indicate that for this context, LLMs offer similar performance in understanding and categorizing complex textual data to traditional machine learning techniques, suggesting that use of LLMs in this context may be unnecessary, and that the trade-off ultimately comes down to the operational costs and resource consumption of each model. This work contributes to the field by providing insights into the capabilities and limitations of current text categorization techniques in the context of rapidly expanding online marketplaces.**

Keywords—E-commerce, Product Categorization, Large language models, Support Vector Machines, Random Forest, Traditional Machine Learning Algorithms

I. INTRODUCTION

Text classification is one of the most investigated topics in machine learning [1]–[4], and with the exponential growth of digital content, particularly in the e-commerce domain, the development of sophisticated automated text categorization systems has become a necessity. These systems are pivotal in enhancing search efficiency, product discoverability, and user experience on online marketplaces [5], [6]. The advent of machine learning (ML) and, more recently, Large Language Models (LLMs) has introduced significant advancements in this area [7], offering promising solutions to the challenges of text classification.

Automated categorization of products in e-commerce not only streamlines the shopping experience by facilitating easier navigation through product listings but also aids in the management of inventory and recommendation systems. Traditional ML algorithms, such as Support Vector Machines (SVMs) and decision trees, have been extensively employed for text classification tasks, demonstrating considerable success in various domains, including document categorization and incident management [8]. However, the complexity and dynamic nature of e-commerce product listings, characterized by diverse categories and rapidly changing inventories, pose unique challenges that require more advanced solutions. [9]

In this study, we examine the efficacy of cutting-edge LLMs relative to conventional machine learning algorithms for product categorization tasks across e-commerce platforms, which typically entail extensive category systems. We evaluated the success of our models by accuracy, precision, recall and F1 metrics.

The paper is divided into multiple sections. A literature review in the second section, followed by our methodology in the third section which we explain the steps we followed in data collection and preparation, as well as model selection and training. In the fourth section, we report our findings and results, and lastly our conclusion in the fifth section.

II. LITERATURE REVIEW

Automated text categorization has become a crucial area of research with applications ranging from document classification to incident management and product categorization. The increasing volume of textual data across various domains necessitates efficient and accurate categorization methods to enhance information retrieval, organization, and processing. This literature review examines related works in the field, highlighting methodologies, challenges, and advancements that pave the way for our investigation into the categorization of products from online e-commerce websites using traditional machine learning algorithms and LLMs.

Apte et al. (1994) [10] laid foundational work in automated text categorization, employing optimized rule-based induction methods to uncover classification patterns within extensive document collections. Their approach demonstrated that machine-generated decision rules could achieve performance comparable to human efforts, thereby reducing the need for extensive human-engineered systems. The study reported a significant improvement in categorization accuracy on the Reuters collection, achieving an 80.5

The automation of the International Patent Classification (IPC) system by Fall et al. [11] addressed the complexities of categorizing patents within a hierarchical taxonomy. The study underscored the challenge of managing the growing number of patent applications and the necessity for precise classification. By developing a reference collection for training and testing, the research explored various machine learning algorithms' effectiveness in categorizing English-language patent documents, offering insights into the adaptability of automated categorization in specialized fields.

Silva et al. (2018) [12] shifted the focus to IT incident management, proposing the use of machine learning, specifically Support Vector Machines (SVMs), to automate the categorization of incident tickets. This approach aimed to enhance operational efficiency by ensuring accurate assignment to resolution groups, thereby improving resolution times and customer satisfaction. The study achieved an accuracy of approximately 89

Huang et al. (2019) [13] tackled hierarchical multi-label text classification (HMTC), a complex challenge where documents are categorized into multiple levels within a hierarchy. Their innovative Hierarchical Attention-based Recurrent Neural Network (HARNN) framework addressed the dependencies between category levels and the association between texts and hierarchical structures, demonstrating improved classification effectiveness and interpretability.

Jahanshahi et al. (2021) [14] focused on multi-level product category classification for grocery items, comparing six text classification models, including traditional and advanced machine learning methods. The study highlighted the importance of dynamic masking of infeasible subcategories and the incorporation of bilingual product titles in improving classification accuracy. The findings indicated that neural network-based models, leveraging advanced techniques, outperformed traditional methods, offering valuable insights for our research.

In our study, we aim to extend these findings by investigating the automatic categorization of products from online e-commerce websites. We will explore the efficacy of traditional machine learning algorithms in comparison to the capabilities of LLMs, focusing on the potential advantages of LLMs in handling the vast and varied textual data in e-commerce. This investigation seeks to contribute to the ongoing discourse on automated text categorization, specifically in the context of the rapidly expanding e-commerce sector.

III. Methodology

In our work we aimed to develop multiple auto-categorization solution for e-commerce platforms. Implementing various models based on traditional machine learning techniques and much more advanced state-of-the-art LLMs.

Our methodology consists of several phases. In this section we are providing a detailed description of our work.

A. Data Collection

We started our work by collecting data that is suitable for our research. We relied on six of the most common e-commerce platforms in Türkiye. By browsing and scraping the listings of various products within these e-commerce platforms, products titles, descriptions and categorical information were collected. At the end of this phase we were able to collect nearly 2.5 Million products in total, with at least 250,000 products from each e-commerce platform.

B. Data Preparation

Following data collection phase, we passed the collected data through a multi-step process to prepare it into a form suitable to train our models.

Firstly, the collected data exhibited heterogeneity in product categorization across the various e-commerce platforms. To address this problem we internally proposed two solutions:

1) **Building a new framework:** This option was considered but deemed impractical due to the complexity of encompassing all product variations.
2) **Adapting an existing framework:** We leveraged the overlap in product listings across the six different platforms. We selected the categorization scheme from the source with the most listed items. Therefore, only data collected from this particular source will be utilized for training and evaluating the target models.

The existing framework employed a hierarchical categorization scheme, where the upper levels represented broad and abstract family of products, while the lower levels delved into more granular and specific details. To strike a balance between overly general and overly specific categories, we opted for the third level of the hierarchy. Consequently, we selected a 1-level categorization scheme that consist of the categories in the third level in the categories hierarchy of the selected e-commerce platform.

Afterward, our focus concentrated on data quality and representativeness of the possible products spectrum, by handling the following problems in the data:

- **Duplicate Removal:** E-commerce platforms often have listings of the same product in multiple categories, potentially hindering model learning. We identified and eliminated duplicate items to ensure data integrity.
- **Category Imbalance:** The number of products per category displayed significant imbalances, while some containing thousands of items others had very few. To mitigate this, categories with less than 1000 items were excluded, resulting in 65 retained categories.

Our data preparation phase was essential for refining our models, focusing on unifying product categorization and enhancing data quality by removing duplicates and addressing category imbalances. This groundwork ensures our model's effectiveness and adaptability for e-commerce platforms.

C. Models Selection & Training

In this phase, we choose the different models we will perform our tests on, and optimized the hyper-parameters of each model to achieve the best possible outcome out of it.

*1) **Models Selection**:* We opted to work with three models in total, two of to be a traditional machine learning models that showed high performance in text-classification tasks and an LLM for the third model.

- **Random Forest (RF)** is an ensemble machine learning method that enhances accuracy and counters overfitting by amalgamating multiple decision trees. Each tree in the "forest" is trained on random data subsets and feature sets, which ensures varied predictions and robust generalization. This algorithm is prized for its adaptability, proficiency with large, complex datasets, and capability to handle missing data effectively [15].

- **Support Vector Machine (SVM)** is a powerful supervised machine learning algorithm used for both classification and regression tasks. It is most commonly used in classification problems. At its core, SVM aims to find the optimal hyperplane (a boundary) that separates different classes in the feature space. The best hyperplane is the one that has the largest margin, meaning it maximizes the distance between the nearest data points of all classes, which are known as support vectors [16].

- **XLM-RoBERTa** is a multilingual adaptation of the RoBERTa model, pre-trained on a vast 2.5TB dataset from CommonCrawl, covering 100 languages. It employs a self-supervised learning approach, using the Masked Language Modeling (MLM) technique, where it predicts randomly masked words in sentences without human-labeled data. This process enables the model to develop a bidirectional understanding of language from the raw text, making it highly effective for extracting features applicable to various downstream tasks, such as text classification in our case, using the learned language representations [17].

*2) **Models Training**:* In the initial step of training phase, we vectorized our data into features utilizing the Term Frequency-Inverse Document Frequency (TF-IDF) technique, with a threshold of 10,000 features. This was followed by the application of a grid-search algorithm to fine-tune the hyperparameters of the Random Forest and Support Vector Machine (SVM) models. Despite that, the default hyperparameters provided by the scikit-learn library emerged as the most effective configuration. The Random Forest model was specifically configured with an ensemble of 500 estimators. For the RoBERTa model, we used the pretrained RoBERTa tokenizer with max token size of 256 to tokenize the data. Finally, we started the fine-tuning process of RoBERTa with the following hyperparameteres:

Hyperparameter	*Value*
Learning rate	2e-5
Learning scheduler	Linear Decay
Weight decay	0.01
Optimizer	AdamW [18]
Batch size	64
Epochs	10

TABLE I: Hyperparameters used for finetuning RoBERTa

IV. Experiments and Results

Table II presents comparative performance metrics for our models. These metrics are key indicators of each model's ability to accurately classify or predict outcomes based on the given data.

The results show that while RoBERTa has the edge with the highest metrics across the board, except for macro recall, SVM's performance is notably close to RoBERTa's in all categories. Specifically, SVM's accuracy is only 1.1% lower, and its macro precision and recall scores are within a range of about 1-2% of RoBERTa's scores. Moreover, SVM slightly beats RoBERTa with macro recall. This proximity in performance suggests that the advantage of using RoBERTa over SVM for this task is marginal. Considering the computational resources and complexity that often accompany models like RoBERTa, the relatively slight improvement in performance metrics may not justify its use over the more straightforward and potentially less resource-intensive SVM in this specific use case.

The RF model, while trailing behind SVM and RoBERTa, still posts respectable scores, with an accuracy of 85.1%, which is reasonably competitive given that it is only 5.5% lower than RoBERTa's. Its precision, both macro and weighted, is above 85%, and the macro F1 score is at 82.8%, which are solid numbers, although they are slightly lower compared to the other two models. This indicates that while RF may not be as precise or effective in this specific use case as SVM or RoBERTa, it still performs adequately and could be considered a viable option, especially in scenarios where interpretability and ease of model understanding are important, or where model training and inference times are critical factors.

	RF	SVM	RoBERTa
Accuracy	0.851	0.895	**0.906**
Precision (macro)	0.865	0.875	**0.889**
Recall (macro)	0.803	**0.879**	0.873
F1 (macro)	0.828	0.876	**0.880**
Precision (weighted)	0.854	0.896	**0.906**
Recall (weighted)	0.851	0.895	**0.906**
F1 (weighted)	0.850	0.895	**0.906**

TABLE II: Evaluation results of the selected models

V. Conclusion

Our work has provided an evaluation of the effectiveness of traditional machine learning algorithms and LLMs in the domain of product categorization for e-commerce platforms. The comparative analysis revealed that LLMs, specifically bert-variants such as the selected XLM-RoBERTa model, while demonstrating the highest performance metrics, do not significantly outperform the traditional SVM model, suggesting that the incremental gains may not justify the additional computational complexity and resource requirements associated with LLMs in this specific context.

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